MA 20205

Section # 2

(EC, PH)

Probability & Statistics

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Random experiments (R)

Ω = all possible outcomes

sample space

S = subset of sample space which are of interest (event)

Problem 1: Assigning probabilities to events.

Approach of relative frequency.

Perform n independent repetitions of R

$$\text{relative freq. of } S = \frac{\#\ S \text{ occurs}}{n} = f_S$$

Let $n \to \infty$

$$P_s = \lim_{n\to\infty} f_s$$

In reality,

[n - repitition of R ; calculate f_s .] Run

variation in f_s .

M - runs .

1^{st} run - $f_s^{(1)}$

2^{nd} run - $f_s^{(2)}$

⋮

i^{th} run - $f_s^{(i)}$

Note that $f_s^{(i)} \in [0,1]$ $\forall\, i = 1, 2, \ldots, M$

mean & std. deviation of $\{f_s^{(1)}, f_s^{(2)}, \ldots, f_s^{(M)}\}$

Pictorially $\rightarrow$ Plot histogram

(WLLN, CLT)

1) R: Rolling a die

$\Omega = \{1, 2, \ldots, 6\}$

$S = \{1, 2, 3\}$

2) R: Tossing a coin until the first head appears

$\Omega = \{H, TH, TTH, TTTH, TTTTH, \ldots\}$

3) R:

$\Omega = [0, 1]$

$0.\boxed{x_1}\, x_2\, x_3\, x_4 \ldots$

$0.\, y_1\, \boxed{y_2}\, y_3\, y_4 \ldots$

$0.\, z_1\, z_2\, \boxed{z_3}\, z_4 \ldots$

$0.\, x_1\, y_2\, z_3 \ldots$